3 Heat Energy

When the temperature of a substance is raised, its atoms and molecules begin to vibrate and release a kind of energy. This energy is known as heat energy or thermal energy. This energy flows from a hot substance to a cold substance.



Do you know?

Heat is the total energy of the molecules in a body. Temperature is the measure of heat in a body.

If we put some ice cubes into water in a glass, water becomes cold. It is because, heat is transferred from water to ice.

Activity 3



Rub your hands together. What do you feel in your hands? Do you feel the heat generated by friction?

Activity 4



Take a small amount of lime powder in a glass. Add some water and stir well. Touch the glass outside. How do you feel?

In both the cases, you can feel the heat. Thus, heat is produced by friction and chemical reactions also. Sun is the primary source of heat energy.

Uses of Heat Energy

- Heat energy obtained from power stations is used to generate electricity.
- Heat energy obtained from petrol and diesel is used to run vehicles.
- We cook food with the help of heat. Heat energy renders the food material soft and easy to digest.
- Hard substances like iron are heated to mould them into different shapes.
- Heat is used to dry clothes and other wet substances.



4 Light Energy



Do you know?

Study of light is known as Optics.

Light is a form of energy which travels as a wave. It contains a particle called photon which are the minute packets of energy. It is the only form of energy visible to human eye. Light does not require any medium to travel. It travels at a speed of 3,00,000 km/s. Sunlight takes 8 minutes and 20 seconds to reach the earth.

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